

CSCI 5512 Project: Bayesian Belief-State Trading Agent for Autonomous Portfolio Management

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May 2026

Abstract

This project develops a Bayesian belief-state trading agent for autonomous portfolio management on the S&P 500 index. The agent treats the market regime as hidden, estimates regime dynamics with a Gaussian Hidden Markov Model (HMM), and maintains a daily probability distribution over latent regimes using Bayesian filtering. These beliefs are then converted into long-only portfolio weights between cash and the S&P 500. The system is evaluated against three non-belief baselines: Buy and Hold, Moving Average Crossover (50/200), and a Single-Regime No-Belief strategy.

The base Bayesian agent is trained on 2015–2020 data and evaluated on 2021–2025 data. It produces the lowest annualized volatility among the main strategies (9.05%) and the lowest maximum drawdown (-15.11%), but it underperforms Buy and Hold in total return because it is conservative and trades frequently. Additional experiments show that naive entropy scaling does not materially improve results, while turnover control and bear-regime defensive allocation are more useful. A validation-selected enhanced bear-defense strategy, chosen on 2021–2022 and evaluated on held-out 2023–2025 data, achieves 60.78% total return, Sharpe ratio 1.227, Sortino ratio 1.573, and maximum drawdown -15.85%. The final result is therefore not that Bayesian trading always maximizes raw return, but that explicit regime beliefs provide a defensible framework for risk-aware allocation and downside protection. Feature ablations also show that Volume degrades regime detection; the best feature subset is Log_Return + Volatility.

1 Introduction

1.1 Problem Statement and Motivation

Financial markets are difficult environments for autonomous decision-making because the true condition of the market is not directly observable. Prices, volume, and volatility are visible, but the underlying market state that generates those observations is latent. A market may behave like a bull market, a bear market, or a sideways market, but the agent only sees noisy indicators of those regimes.

This project studies whether a trading agent benefits from explicitly modeling that uncertainty. Instead of assuming that the market has one fixed return distribution, the proposed agent maintains a probability distribution over hidden regimes. Each day, it updates this belief using new observations and then adjusts its portfolio exposure based on expected return and risk under the current belief. The setting is naturally connected to stochastic artificial intelligence because the agent must make sequential decisions under partial observability.

The practical motivation is risk management. A passive strategy such as Buy and Hold can perform very well in long upward markets, but it has no mechanism for reducing exposure when market conditions become dangerous. A belief-state strategy may give up some upside in strong bull periods, but it can potentially reduce volatility, drawdown, and exposure during uncertain or crisis-like periods. The project therefore

evaluates both return and risk-aware behavior rather than only asking which strategy has the highest total return.

1.2 Problem Formulation

The trading problem is formulated as a partially observable decision process. The hidden state is the market regime, the observation is daily market data, and the action is the fraction of capital invested in the S&P 500. The remaining capital is held in cash. The action is restricted to a long-only interval:

$$w_t \in [0, 1].$$

The agent maintains a belief state

$$b_t(s) = P(s_t = s \mid o_{1:t}),$$

where s_t is the hidden regime at time t and $o_{1:t}$ is the sequence of observations through day t . The agent updates this belief recursively, computes expected risk and return under the belief, and chooses a portfolio weight. Transaction costs are included to make the comparison more realistic.

1.3 Project Contributions

The project contributes an end-to-end implementation and evaluation pipeline:

- a reproducible S&P 500 data loading and preprocessing pipeline;
- three non-belief baselines for comparison;
- Gaussian HMM regime estimation with multiple random restarts;
- Bayesian filtering over hidden market regimes;
- mean-variance and belief-conditioned portfolio decisions;
- fair turnover-based transaction cost handling;
- evaluation metrics including return, volatility, Sharpe, Sortino, maximum drawdown, Calmar ratio, belief calibration, turnover, and entropy;
- HMM model-selection, feature, risk-aversion, and entropy ablations;
- a pgmpy Dynamic Bayesian Network representation of the same belief-state structure;
- an enhanced bear-defense strategy with validation and held-out testing.

2 Related Work

Mean-Variance Portfolio Optimization. The project starts from the classical mean-variance portfolio framework of Markowitz [1]. In that framework, an investor balances expected return against variance. The challenge in this project is that expected return and variance are not stable constants: they depend on the hidden market regime and must be estimated from data.

Regime-Switching Financial Models. Hamilton’s work on Markov-switching models provides a foundational approach for modeling discrete latent regimes in time series [2]. Hidden Markov Models are a natural fit for market regimes because they combine a latent Markov chain with an observation model. Rabiner’s tutorial gives the standard estimation and inference background for HMMs [3]. In financial settings, Tu [4] shows that accounting for bull and bear regimes can matter for portfolio decisions, and Nystrup, Madsen, and Lindstrom [5] show that hidden-regime information can be used in dynamic portfolio allocation with transaction costs.

Bayesian Beliefs and POMDPs. Partially Observable Markov Decision Processes formalize decision-making when the agent cannot directly observe the state of the environment [7]. In a POMDP, the belief state is a sufficient statistic for action selection. Bayesian reinforcement learning extends this idea by treating uncertainty itself as part of the agent’s knowledge state [8]. This project uses a simplified version of that idea: the HMM parameters are learned offline, and the agent performs online Bayesian filtering over regimes during the test period.

Dynamic Bayesian Networks for Trading. Dynamic Bayesian Networks provide a graphical-model view of temporal hidden-state systems. O et al. [6] use DBN-style hidden-state modeling for stock-trading signals. In this project, pgmpy is used to represent the regime model as a Dynamic Bayesian Network. The final trading filter remains HMM-based because the observations are continuous Gaussian features, but the DBN prototype helps connect the implementation to the Bayesian network language discussed in the project plan.

Positioning of This Project. Prior work motivates the individual components of the system: mean-variance allocation, hidden market regimes, Bayesian filtering, and dynamic Bayesian networks. The contribution of this project is not a new financial theorem or a production trading system. Instead, the project combines these ideas into one reproducible educational pipeline and evaluates the value of explicit regime beliefs against non-belief baselines on the same S&P 500 test period. The most important comparison is therefore not only raw return, but whether belief tracking changes risk behavior, drawdown, calibration, turnover, and performance during unstable periods.

3 Methods and Algorithms

3.1 End-to-End Workflow

Figure 1 summarizes the implemented pipeline. The figure is intentionally written as a simple report-native flowchart rather than a separate diagram dependency.

3.2 POMDP-Style Formulation

The project can be described using a POMDP-style tuple:

- **States S :** latent market regimes, represented by $K = 3$ hidden states.
- **Actions A :** portfolio weights $w_t \in [0, 1]$ invested in the S&P 500.
- **Observations O :** daily features such as log return, volume, and volatility.
- **Transition model T :** HMM transition probabilities $P(s_t | s_{t-1})$.
- **Observation model:** Gaussian likelihood $P(o_t | s_t)$ for each regime.
- **Reward:** realized portfolio return after transaction costs, interpreted through risk-adjusted metrics.

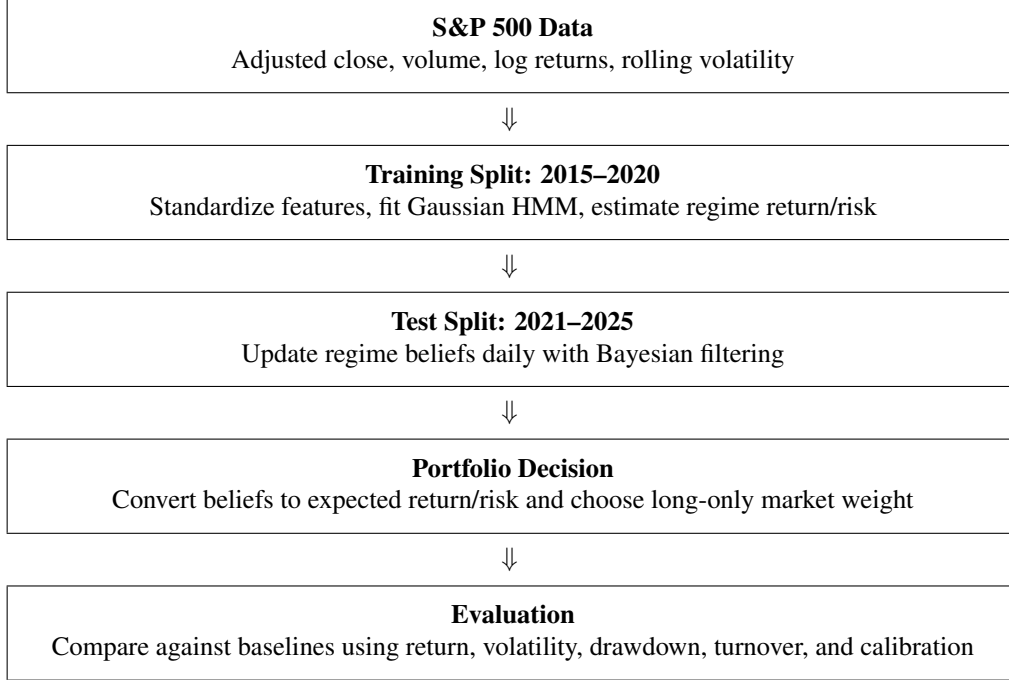


Figure 1: High-level workflow of the belief-state trading pipeline.

3.3 HMM Parameter Estimation

The main regime model is a Gaussian HMM trained with `hmmlearn`. The main pipeline uses three standardized features: `Log_Return`, `Volume`, and `Volatility`. The HMM learns initial state probabilities, transition probabilities, state-specific Gaussian means, and state-specific covariance matrices.

Because the HMM likelihood surface can be sensitive to initialization, the implementation fits 10 random restarts and keeps the model with the highest training log-likelihood. The HMM states are not learned with semantic labels such as “bull” or “bear.” They are numerical states, and their financial interpretation is assigned after training by inspecting each state’s realized return and volatility on the training split.

Algorithm 1 Belief-State Trading Agent

- 1: Fit Gaussian HMM on training observations
 - 2: Estimate return mean and variance for each decoded training regime
 - 3: Initialize b_0 over hidden regimes
 - 4: **for** each test day t **do**
 - 5: Observe feature vector o_t
 - 6: Predict prior belief using transition matrix T
 - 7: Compute Gaussian likelihood $P(o_t | s)$ for each regime
 - 8: Update and normalize posterior belief b_t
 - 9: Compute predictive return mean and variance from b_t
 - 10: Choose portfolio weight $w_t \in [0, 1]$
 - 11: Apply turnover-based transaction cost
 - 12: Record daily return and diagnostics
 - 13: **end for**
-

3.4 Bayesian Filtering

The belief update follows the standard forward-filtering equation:

$$b_t(s) \propto P(o_t | s) \sum_{s'} P(s | s') b_{t-1}(s').$$

The summation term is the prediction step, and the likelihood term is the correction step. The implementation performs the calculation in log space for numerical stability. This matters because Gaussian likelihoods can become very small over long time series.

3.5 Portfolio Decision Rule

For each regime s , the training data gives an estimated daily return mean μ_s and variance σ_s^2 . Given belief b_t , the predictive mean is

$$\bar{\mu}_t = \sum_s b_t(s) \mu_s.$$

The implementation uses the law of total variance:

$$\bar{\sigma}_t^2 = \sum_s b_t(s) \sigma_s^2 + \sum_s b_t(s) (\mu_s - \bar{\mu}_t)^2.$$

The first term is the average within-regime variance. The second term captures uncertainty from not knowing which regime is active. This second term is important because it makes the agent more cautious when belief mass is spread across states with different expected returns.

The base portfolio weight is a long-only mean-variance rule:

$$w_t = \frac{\bar{\mu}_t}{2\lambda\bar{\sigma}_t^2}, \quad w_t \leftarrow \min(1, \max(0, w_t)).$$

The base strategy uses $\lambda = 2.0$.

3.6 Belief Entropy

The project plan discusses exploration and exploitation through belief uncertainty. The implementation measures uncertainty with Shannon entropy:

$$H(b_t) = - \sum_s b_t(s) \log b_t(s),$$

and normalized entropy:

$$H_{\text{norm}}(b_t) = \frac{H(b_t)}{\log K}.$$

Low entropy means the agent is confident in one regime; high entropy means belief is spread across multiple regimes. Entropy is reported as a diagnostic and tested as a portfolio-scaling signal.

3.7 Enhanced Bear-Defense Strategy

The base Bayesian strategy is useful for risk control, but it changes weights frequently and pays high transaction costs. The enhanced strategy uses the belief state more selectively. It fits the HMM using the feature-ablation-informed feature set Log_Return and Volatility (i.e., it refits the HMM without the Volume

channel, motivated by Table 10), then stays mostly invested unless the posterior probability of the bear-like state becomes elevated. The final validation-selected rule is:

$$w_t = \max(w_{\min}, 1 - \gamma \max(0, b_t(\text{bear}) - \tau)).$$

The selected configuration uses $\tau = 0.25$, $\gamma = 3.0$, $w_{\min} = 0.0$, smoothing parameter 0.35, and rebalancing threshold 0.03. This rule is not claimed to be globally optimal; it is the strongest validated improvement found in the current project pipeline.

3.8 Baselines

Three baselines are included:

- **Buy and Hold:** always fully invested in the S&P 500.
- **Moving Average Crossover 50/200:** invested when the 50-day moving average is above the 200-day moving average; otherwise in cash.
- **Single-Regime No-Belief:** estimates one stationary return distribution from training data and applies one fixed mean-variance weight.

The Single-Regime baseline is especially important because it uses the same general portfolio logic as the Bayesian agent, but without regime beliefs.

4 Experimental Setup

4.1 Data and Features

The project uses daily S&P 500 data from Yahoo Finance. The adjusted close price is used for backtesting returns to avoid distortions from splits and dividends. The main HMM features are:

- Log_Return,
- Volume,
- rolling Volatility.

Features are standardized using training-set statistics only, preventing leakage from the test period.

Split	Date Range	Rows	Purpose
Training	2015-01-02 to 2020-12-31	1511	HMM fitting and return/risk estimates
Test	2021-01-04 to 2025-12-30	1254	belief updates, trading, evaluation

Table 1: Training and test split used by the main pipeline.

For trading evaluation, the code separately computes real daily log returns from adjusted close:

$$\text{Real_Log_Return}_t = \log \left(\frac{\text{Adj_Close}_t}{\text{Adj_Close}_{t-1}} \right).$$

This keeps standardized model features separate from the actual returns used in the backtest.

4.2 Validation for the Enhanced Strategy

The main HMM and base strategies are evaluated on 2021–2025. For the enhanced bear-defense strategy, the project adds a more conservative validation check:

- HMM training: 2015–2020;
- enhanced configuration selection: 2021–2022 validation window;
- held-out evaluation: 2023–2025.

This split reduces the risk of choosing a portfolio rule only because it performs well on the full test period.

4.3 Transaction Costs

The project applies a transaction cost proportional to turnover:

$$\text{cost}_t = c|w_t - w_{t-1}|,$$

where $c = 0.001$, corresponding to 0.1% or 10 basis points. This cost is applied consistently to strategies that change exposure. This is important because the original Bayesian strategy trades more often than the baselines.

4.4 Evaluation Metrics

The report uses the following metrics:

- total return and annualized return;
- annualized volatility;
- Sharpe ratio;
- Sortino ratio;
- maximum drawdown;
- Calmar ratio;
- average portfolio weight;
- turnover and transaction cost;
- belief calibration and entropy.

The Calmar ratio is annualized return divided by the absolute value of maximum drawdown; higher is better. These metrics are chosen because the project is about risk-aware trading, not only raw return maximization.

4.5 Reproducibility

The code is organized into reusable modules in `src/` and numbered scripts in `scripts/`. The team repository is available on GitHub at [rowe0137/5512-Final-Stock-Trader](https://github.com/rowe0137/5512-Final-Stock-Trader). From the `belief_state_trader/` directory, the full pipeline is regenerated with:

```
python scripts/run_all.py.
```

This command regenerates the result CSV files and plots used in this report.

5 Results

5.1 Learned HMM Regimes

The fitted $K = 3$ HMM learns three interpretable states. Table 2 shows the training-period interpretation. The HMM itself only learns state indices; bear, sideways, and bull are post-hoc labels assigned from return and volatility statistics.

State	Frequency	Annual Return	Annual Volatility	Interpretation
state_0	4.0%	-70.87%	64.71%	Bear-like / high-risk
state_1	31.3%	6.86%	20.69%	Sideways / moderate
state_2	64.7%	16.55%	9.06%	Bull-like / lower-risk

Table 2: HMM state interpretation from the training split.

The rare bear-like state is extreme because the training period includes the March 2020 COVID crash. This affects later regime interpretation: a milder decline such as the 2022 bear market does not always look like the extreme state learned from 2020.

5.2 Model Selection

The model-selection sweep compares $K = 2$ through $K = 8$. Table 3 and Figure 2 show that AIC and BIC continue decreasing as K increases. Therefore, $K = 3$ is not presented as statistically optimal under AIC/BIC. It is kept because it is interpretable and directly aligned with the proposal’s bull, sideways, and bear regime framing.

K	Log-Likelihood	AIC	BIC
2	-4169.13	8380.27	8492.00
3	-3497.48	7064.95	7251.17
4	-2928.26	5958.52	6229.87
5	-2600.40	5338.80	5705.92
6	-2396.69	4971.38	5444.91
7	-2219.84	4661.68	5252.26
8	-2049.79	4369.58	5087.85

Table 3: HMM model-selection sweep. Lower AIC/BIC values favor larger models in this range; the main report keeps $K = 3$ for interpretability.

5.3 Base Strategy Comparison

Table 4 reports the main strategy comparison for the 2021–2025 test period. Figure 3 shows the equity curves, and Figure 4 shows the drawdown behavior.

The base Bayesian agent has the lowest annualized volatility and lowest maximum drawdown among the main strategies. This supports the idea that belief-state modeling can make the strategy more risk-aware. However, it has the lowest total return because it is conservative and trades frequently. The result is consistent with the test-period context: 2021–2025 is largely upward-trending, so being underinvested is costly.

The Single-Regime No-Belief strategy has the same Sharpe and Sortino ratios as Buy and Hold because it is essentially a constant scaling of the same return stream. The smaller fixed exposure lowers drawdown

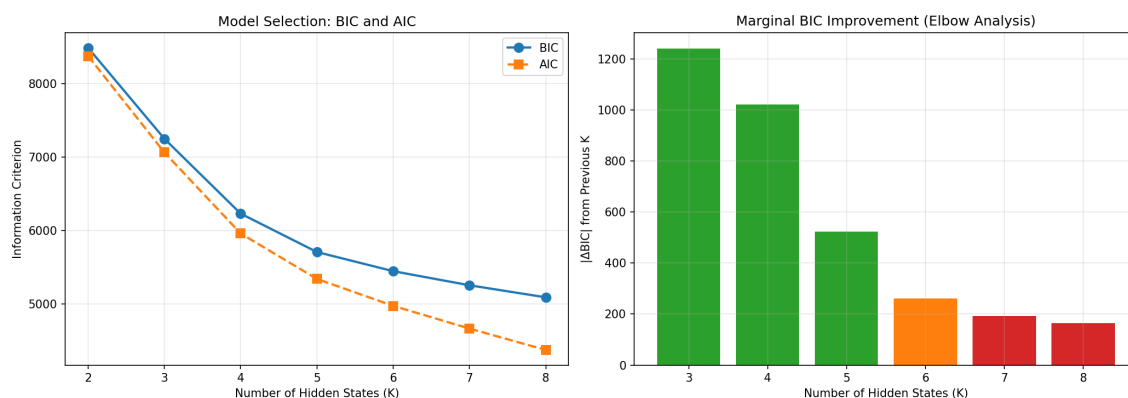


Figure 2: AIC/BIC and log-likelihood trends across hidden-state counts.

Strategy	Total Return	Ann. Return	Ann. Vol.	Sharpe	Sortino	Max DD	Avg Weight
Buy and Hold	85.04%	12.36%	16.95%	0.729	0.983	-25.43%	100.00%
Moving Average 50/200	33.19%	5.76%	12.07%	0.477	0.499	-19.90%	62.09%
Single-Regime No-Belief	55.12%	8.82%	12.09%	0.729	0.983	-18.88%	71.34%
Bayesian Belief-State	21.57%	3.91%	9.05%	0.432	0.561	-15.11%	61.10%

Table 4: Main strategy comparison on the 2021–2025 test period.

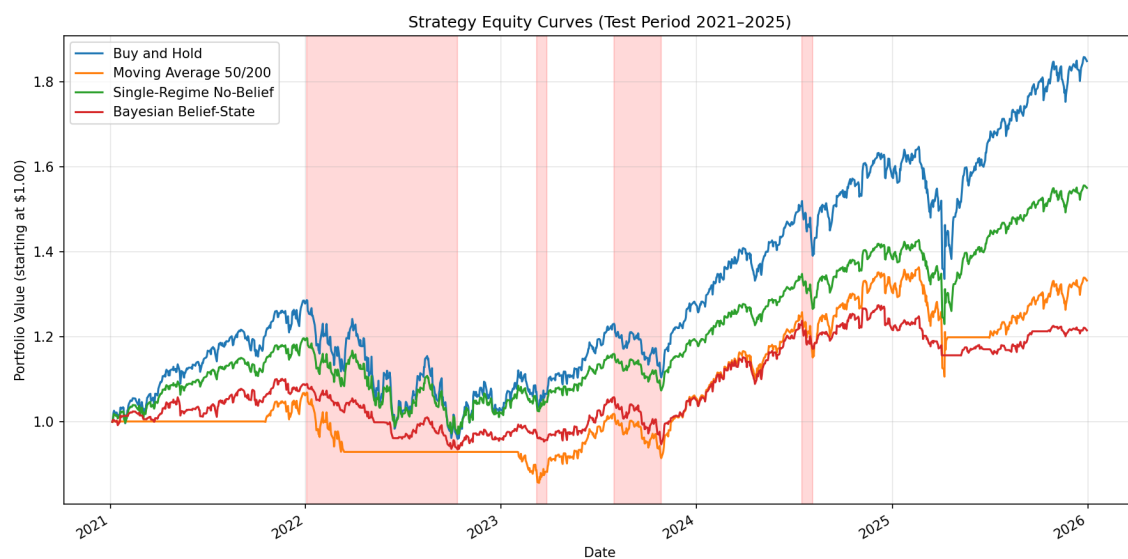


Figure 3: Equity curves for the base strategy comparison.

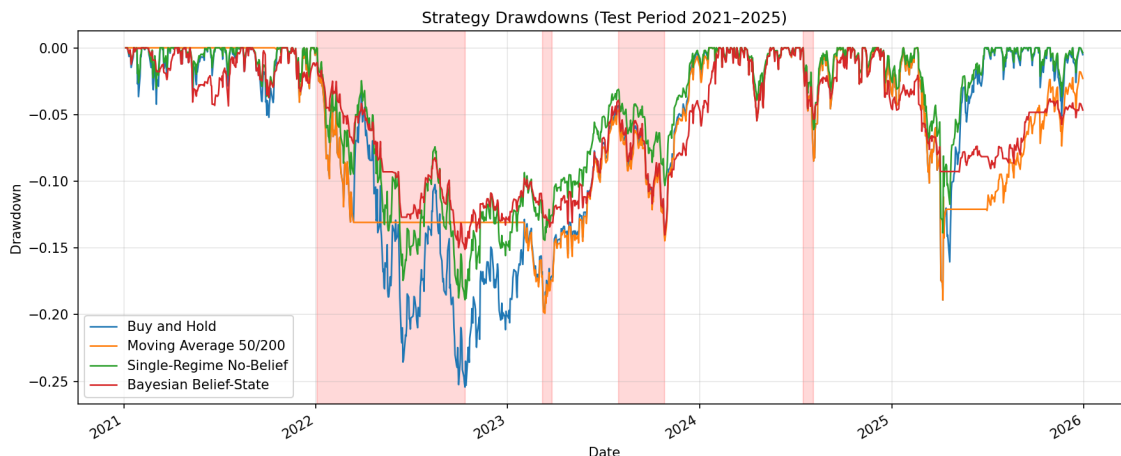


Figure 4: Drawdown comparison for the base strategies.

but does not change the mean-to-volatility ratio. This is a useful reminder that Sharpe ratio alone does not fully describe position-sizing behavior.

5.4 Belief Probabilities and Calibration

Figure 5 shows the daily belief probabilities over the three hidden regimes. The beliefs are often concentrated in one state, but they shift during volatile periods.

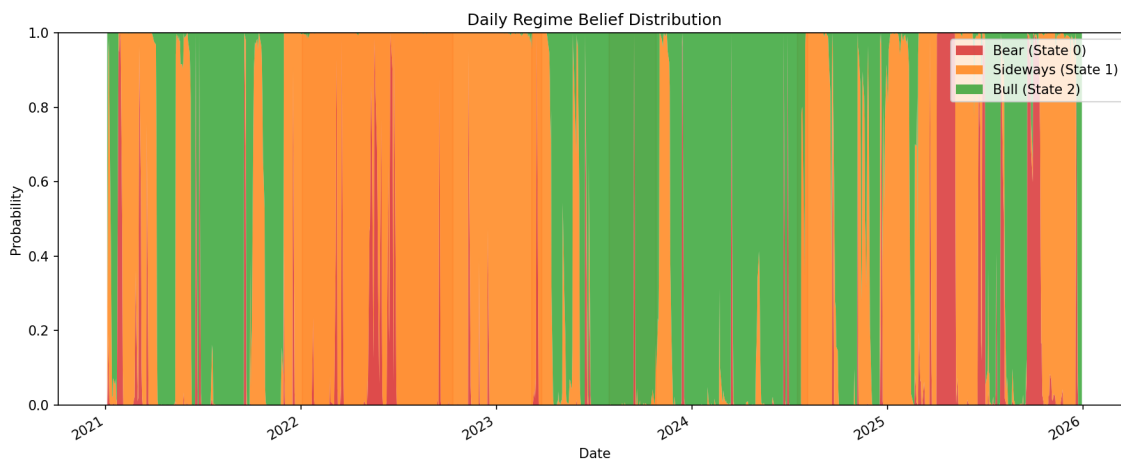


Figure 5: Daily posterior regime beliefs produced by Bayesian filtering.

The belief calibration check asks whether predictive intervals contain realized returns at approximately the expected rates. Table 5 shows that the intervals are slightly conservative but close to the intended coverage levels, especially at the 90% and 95% levels.

The 91.54% empirical coverage at the 90% target is one of the strongest validation results in the project. It means that when the agent's belief-weighted predictive distribution says a realized return should fall inside a 90% interval, the realized return does so at almost the nominal rate. This makes the beliefs more than a black-box signal: they produce uncertainty estimates that are reasonably trustworthy for downstream risk

Target Coverage	Actual Coverage	Calibration Error
50%	57.46%	+7.46%
75%	78.85%	+3.85%
90%	91.54%	+1.54%
95%	96.09%	+1.09%

Table 5: Belief-weighted predictive interval calibration.

decisions. The slight over-coverage at lower confidence levels reflects conservative intervals when regime beliefs are mixed, which is a safer error than overconfidence in a financial setting.

5.5 Performance During Stress Periods

The base Bayesian agent is most useful during difficult periods. Table 6 summarizes selected stress windows. During normal periods the Bayesian agent gives up substantial upside because it is only partially invested, but it also keeps drawdown lower. During the 2022 inflation and rate-hike bear market, Buy and Hold loses 24.95%, while the Bayesian agent loses 13.94%. During the 2024 August volatility event, the Bayesian agent also loses less than Buy and Hold. The Single-Regime baseline is included to separate the value of lower exposure from the value of time-varying belief updates.

Period	Strategy	Total Return	Sharpe	Max Drawdown
Normal periods	Buy and Hold	198.30%	1.887	-18.90%
Normal periods	Single-Regime No-Belief	118.08%	1.887	-13.88%
Normal periods	Bayesian Belief-State	68.36%	1.552	-9.31%
2022 Bear Market	Buy and Hold	-24.95%	-1.529	-25.43%
2022 Bear Market	Single-Regime No-Belief	-18.52%	-1.529	-18.88%
2022 Bear Market	Bayesian Belief-State	-13.94%	-2.334	-14.16%
2024 August Volatility	Buy and Hold	-7.90%	-6.592	-8.49%
2024 August Volatility	Single-Regime No-Belief	-5.70%	-6.592	-6.13%
2024 August Volatility	Bayesian Belief-State	-4.76%	-5.126	-5.37%

Table 6: Selected period-level performance. The Bayesian strategy reduces losses in major stress windows but sacrifices upside in normal periods.

The mechanism is also important. The Bayesian agent does not need to perfectly predict a crash in advance. When observations stop looking confidently bullish, posterior probability shifts toward the sideways or bear-like regimes. At the same time, the predictive variance increases through the law of total variance because the belief state is spread across regimes with different return profiles. The mean-variance optimizer then cuts exposure automatically. This explains why the strategy can reduce losses during transition periods even when the hidden state labels are imperfect.

5.6 Entropy and Ablation Findings

Entropy is theoretically attractive because it measures uncertainty in the belief state. In practice, the entropy ablation shows that directly penalizing entropy reduces return without materially improving maximum drawdown. Table 7 summarizes the result.

The entropy diagnostics are still useful even though direct entropy scaling is not the best trading rule. Table 8 shows that entropy measures regime ambiguity, not simply whether the market is rising or falling. Slow, persistent declines such as the 2022 bear market can have lower entropy because the observations

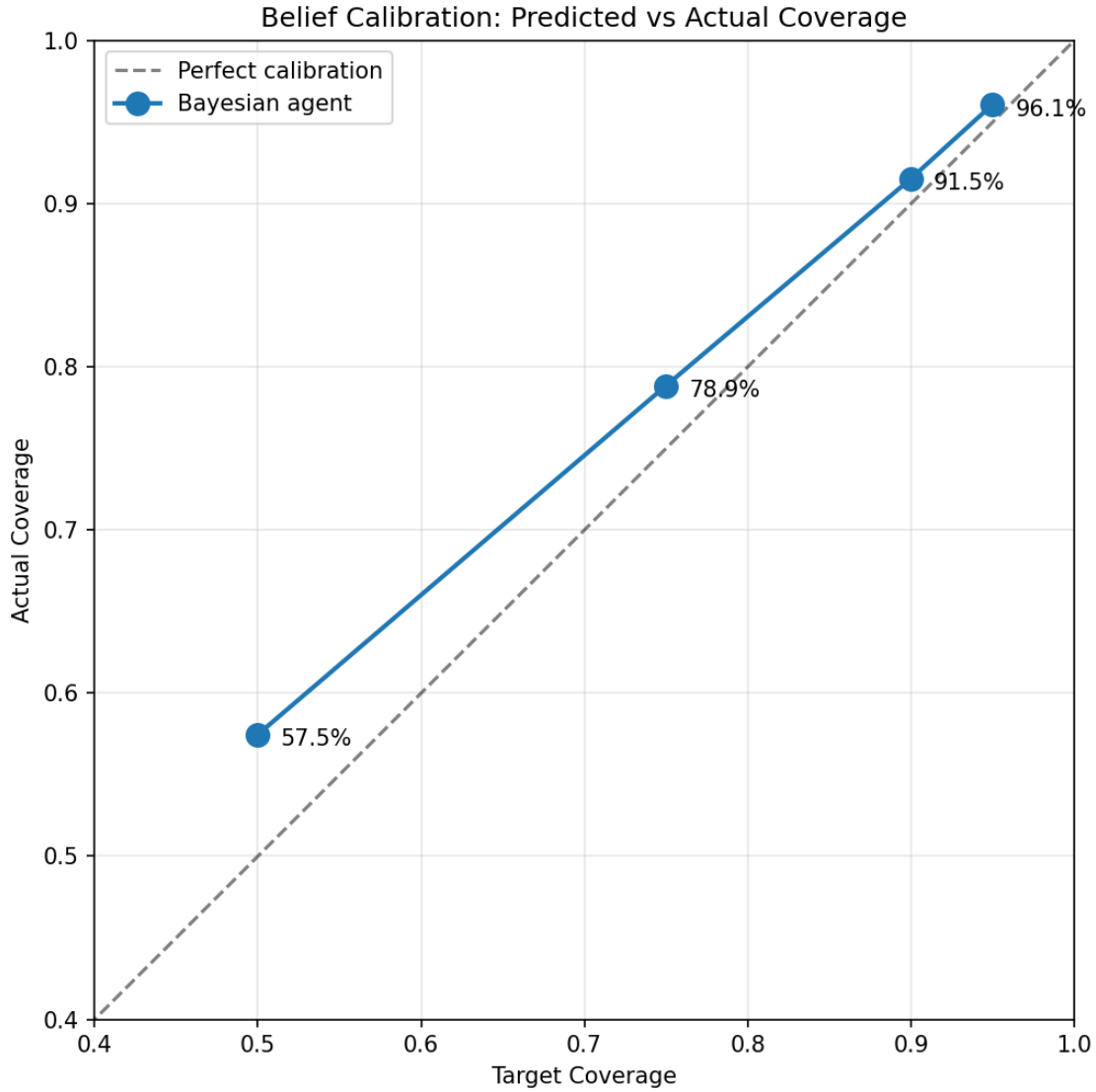


Figure 6: Calibration plot for predictive return intervals.

Method	Total Return	Sharpe	Max DD	Avg Weight
No entropy scaling	21.57%	0.432	-15.11%	61.10%
Penalty 0.5	15.28%	0.327	-15.06%	58.30%
Penalty 1.0	10.57%	0.237	-15.00%	56.19%
Penalty 2.0	3.98%	0.095	-14.87%	53.27%
Penalty 3.0	0.27%	0.006	-15.37%	51.32%

Table 7: Entropy scaling ablation. Direct entropy penalties reduce exposure and return without meaningfully improving drawdown.

consistently match one learned state. Sudden events, such as the August 2024 volatility shock, have higher entropy because the observations are more ambiguous across regimes.

Period	Days	Mean Entropy	Mean Normalized Entropy
Normal periods	965	0.116	0.106
2022 Bear Market	196	0.092	0.084
2023 Banking Crisis	14	0.108	0.098
2023 Q3 Correction	64	0.039	0.036
2024 August Volatility	15	0.208	0.190

Table 8: Belief entropy by market period. Entropy is highest during sudden regime ambiguity rather than during every negative-return period.



Figure 7: Belief entropy and portfolio weights over time.

Risk-aversion tuning provides a clearer improvement than direct entropy scaling. Table 9 shows the base Bayesian strategy under different λ values. Lower λ values increase market exposure but also increase drawdown. Higher λ values reduce drawdown but eventually become too conservative. In this experiment, $\lambda = 3.0$ gives the strongest Sharpe and Sortino ratios for the base mean-variance rule, while $\lambda = 2.0$ remains the main reported base setting from the original pipeline.

Feature ablations show that Volume is not helpful in the current setup. The Log_Return + Volatility feature set produces stronger risk-adjusted performance than the full three-feature set for the base mean-variance strategy. This finding motivates the enhanced strategy's feature choice.

λ	Total Return	Sharpe	Sortino	Max DD	Avg Weight
0.5	22.02%	0.271	0.343	-33.03%	87.60%
1.0	22.90%	0.331	0.433	-26.31%	78.67%
1.5	22.48%	0.403	0.533	-18.79%	67.16%
2.0	21.57%	0.432	0.561	-15.11%	61.10%
3.0	20.83%	0.460	0.575	-11.93%	54.76%
5.0	16.56%	0.403	0.473	-10.52%	48.88%
10.0	8.18%	0.224	0.247	-10.09%	42.35%

Table 9: Risk-aversion sweep for the base Bayesian mean-variance strategy.

Features	Total Return	Sharpe	Max DD	Avg Weight
Log_Return only	26.12%	0.447	-12.51%	70.38%
Log_Return + Volume	-5.34%	-0.154	-17.75%	33.61%
Log_Return + Volatility	22.15%	0.552	-8.54%	44.98%
All three features	21.57%	0.432	-15.11%	61.10%

Table 10: Feature ablation for the Bayesian strategy.

5.7 Enhanced Bear-Defense Strategy

The strongest practical improvement is the enhanced bear-defense rule. The full-period enhanced sweep gives a descriptive result of 70.59% total return, Sharpe 0.771, Sortino 0.987, and maximum drawdown -20.56% over 2021–2025. However, because that sweep uses the full period for exploration, the more defensible result is the validation-selected held-out check shown in Table 11.

Strategy	Total Return	Ann. Return	Ann. Vol.	Sharpe	Sortino	Max DD	Avg Weight	Cost
Buy and Hold	79.61%	19.65%	15.03%	1.308	1.726	-18.90%	100.00%	0.00%
Moving Average 50/200	43.48%	12.11%	14.10%	0.859	1.057	-18.90%	90.15%	0.30%
Single-Regime No-Belief	51.86%	14.02%	10.72%	1.308	1.726	-13.88%	71.34%	0.00%
Enhanced Bear-Defense Bayesian	60.78%	15.80%	12.88%	1.227	1.573	-15.85%	94.72%	0.20%

Table 11: Held-out 2023–2025 results after selecting the enhanced configuration on 2021–2022 validation data.

The enhanced strategy also addresses the transaction-cost problem observed in the original Bayesian agent. Table 12 reports the full-period exploratory turnover diagnostics. The original Bayesian agent changes exposure frequently, producing high turnover and a large cost drag. The enhanced bear-defense rule trades much less often because it stays near full exposure unless the bear-state belief becomes elevated.

The enhanced strategy does not beat Buy and Hold on total return in the held-out 2023–2025 window, which is again a strong upward period. It does, however, improve substantially over the original Bayesian agent by staying invested most of the time and paying much lower transaction costs. The result supports the project goal in a careful way: beliefs are most useful as a risk-control signal, not as a guarantee of maximum raw return.

Strategy	Total Turnover	Trades	Cost Drag	Avg Weight
Buy and Hold	0.00	0	0.00%	100.00%
Single-Regime No-Belief	0.00	0	0.00%	71.34%
Original Bayesian	81.73	656	8.17%	61.10%
Enhanced Bayesian	8.34	60	0.83%	89.41%

Table 12: Full-period exploratory turnover and transaction-cost diagnostics. The enhanced strategy sharply reduces daily rebalancing compared with the original Bayesian agent.

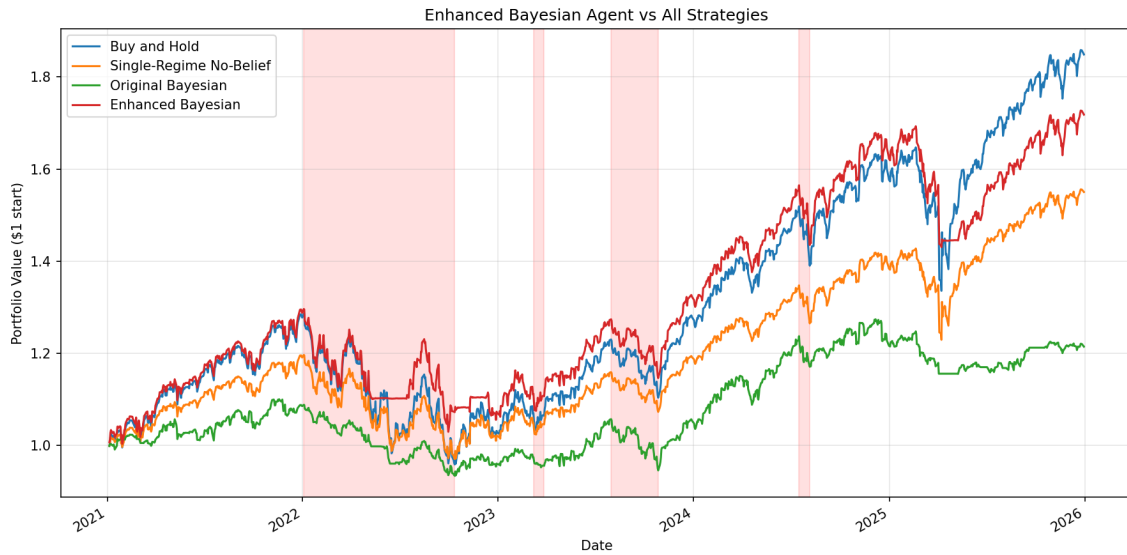


Figure 8: Full-period enhanced strategy equity-curve comparison. The full-period sweep was used for exploration only; Table 11 reports the unbiased held-out 2023–2025 evaluation.

5.8 pgmpy Dynamic Bayesian Network Check

The pgmpy component represents the regime process as a Dynamic Bayesian Network with a temporal regime edge and observed feature nodes. Because the continuous Gaussian likelihoods are learned by the HMM, the DBN result is best interpreted as a representation check rather than a replacement model.

The DBN has two time slices. Within a slice, the hidden node Regime_t points to the observed feature nodes Log_Return_t , Volume_t , and Volatility_t . Across slices, the temporal edge $\text{Regime}_t \rightarrow \text{Regime}_{t+1}$ encodes the Markov transition. Forward inference in this DBN with the same Gaussian conditional distributions is mathematically equivalent to the HMM forward filter, so the pgmpy version mainly makes the graphical-model structure explicit.

Using the same transition and Gaussian emission parameters, the DBN-style forward update agrees with the HMM belief filter on 99.92% of most-likely regime labels. The mean absolute belief-probability difference is 0.0006. This high agreement is expected because the two methods are implementing the same Bayesian filtering idea with the same learned parameters. The DBN serves as a structural representation check, not as independent empirical evidence.

6 Discussion

6.1 Main Findings

The project produces three main findings.

First, the base Bayesian belief-state agent is risk-aware. It lowers volatility and maximum drawdown relative to the baselines, but it is too conservative and suffers from high turnover costs. This shows that belief tracking helps with risk control but does not automatically produce high returns.

Second, entropy is useful as a diagnostic, but direct entropy-based exposure scaling is not the strongest improvement in this test period. The law-of-total-variance portfolio rule already makes the agent more cautious when regime beliefs are uncertain.

Third, the enhanced bear-defense strategy is a better practical use of the belief state. It stays invested by default and reduces exposure only when posterior bear probability rises. This preserves much more upside while still using Bayesian beliefs for risk control.

6.2 Alignment with the Project Plan

The final implementation is aligned with the proposal. It includes hidden regimes, HMM parameter estimation, Bayesian belief updates, belief-conditioned portfolio decisions, transaction costs, baselines, and out-of-sample evaluation. The pgmpy work also addresses the team note about Bayesian network libraries. The main difference from the initial plan is that the final project is more careful about validation: the enhanced strategy is reported with a 2021–2022 validation split and a 2023–2025 held-out check.

6.3 Limitations

Several limitations remain:

- **Single asset:** the agent trades only the S&P 500 against cash. A multi-asset version could allocate across equities, bonds, commodities, or sectors.
- **Fixed HMM parameters:** the HMM is trained once on 2015–2020 and is not updated online.
- **Memoryless regimes:** the HMM transition model does not depend on how long the market has been in a regime.

- **K selection:** higher K values improve AIC/BIC, so $K = 3$ should be viewed as interpretable rather than statistically best. The marginal AIC/BIC improvement does not flatten in the swept range, so a fully data-driven choice would prefer $K \geq 8$.
- **COVID training effect:** the training-period COVID crash makes the bear-like state very extreme, so milder bear markets may not trigger that state strongly.
- **Enhanced feature set:** the enhanced strategy uses Log_Return + Volatility, not the full three-feature HMM. This is justified by ablation results but should be disclosed clearly.
- **Limited robustness evidence:** the held-out 2023–2025 evaluation provides a useful check against tuning directly on the full test period, but it is still only one market history. A stronger robustness study would use rolling train-validation-test windows across multiple historical periods and report uncertainty bands or confidence intervals for the performance metrics.

6.4 Ethical Considerations

Automated trading systems can influence market behavior if deployed at scale. If many algorithms respond to similar regime signals, they may amplify selling during stress periods. Backtests can also create false confidence, especially when a strategy is tuned on a limited historical period. For these reasons, the results should be interpreted as a research prototype, not as investment advice. Any real deployment would require additional validation, risk controls, and transparency.

7 Conclusion

This project implements a complete Bayesian belief-state trading pipeline. The final system estimates hidden market regimes with an HMM, updates posterior regime beliefs through Bayesian filtering, and uses those beliefs to make risk-aware portfolio decisions. The base Bayesian agent reduces volatility and drawdown but gives up return because it is conservative and expensive to rebalance. The enhanced bear-defense strategy shows a more practical use of the belief state: remain invested most of the time, but reduce exposure when posterior bear-regime probability becomes elevated.

The final conclusion is nuanced. In a bull-heavy period, Buy and Hold remains difficult to beat on total return. However, belief-state modeling provides a principled way to represent uncertainty, evaluate risk, and reduce downside exposure. The project therefore supports the value of Bayesian regime beliefs for risk-aware autonomous trading, while also showing that portfolio design and transaction-cost control are just as important as the regime model itself.

Reproducibility Statement

All report tables and figures are generated from the local project pipeline in `belief_state_trader/`. The project repository is available on GitHub at [rowe0137/5512-Final-Stock-Trader](https://github.com/rowe0137/5512-Final-Stock-Trader). The main command is:

```
python scripts/run_all.py.
```

The report source is stored in `belief_state_trader/reports/`, and generated result files are stored in `belief_state_trader/results/`.

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